Aakar — Backend Architecture (v1)

> Two FastAPI services, one shared philosophy: every row is Mandala-scoped, every Vidya is registered, every Yajna is auditable. This document covers the request lifecycle, the database, the Yajna lifecycle, and the operational concerns.

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1. Two backends

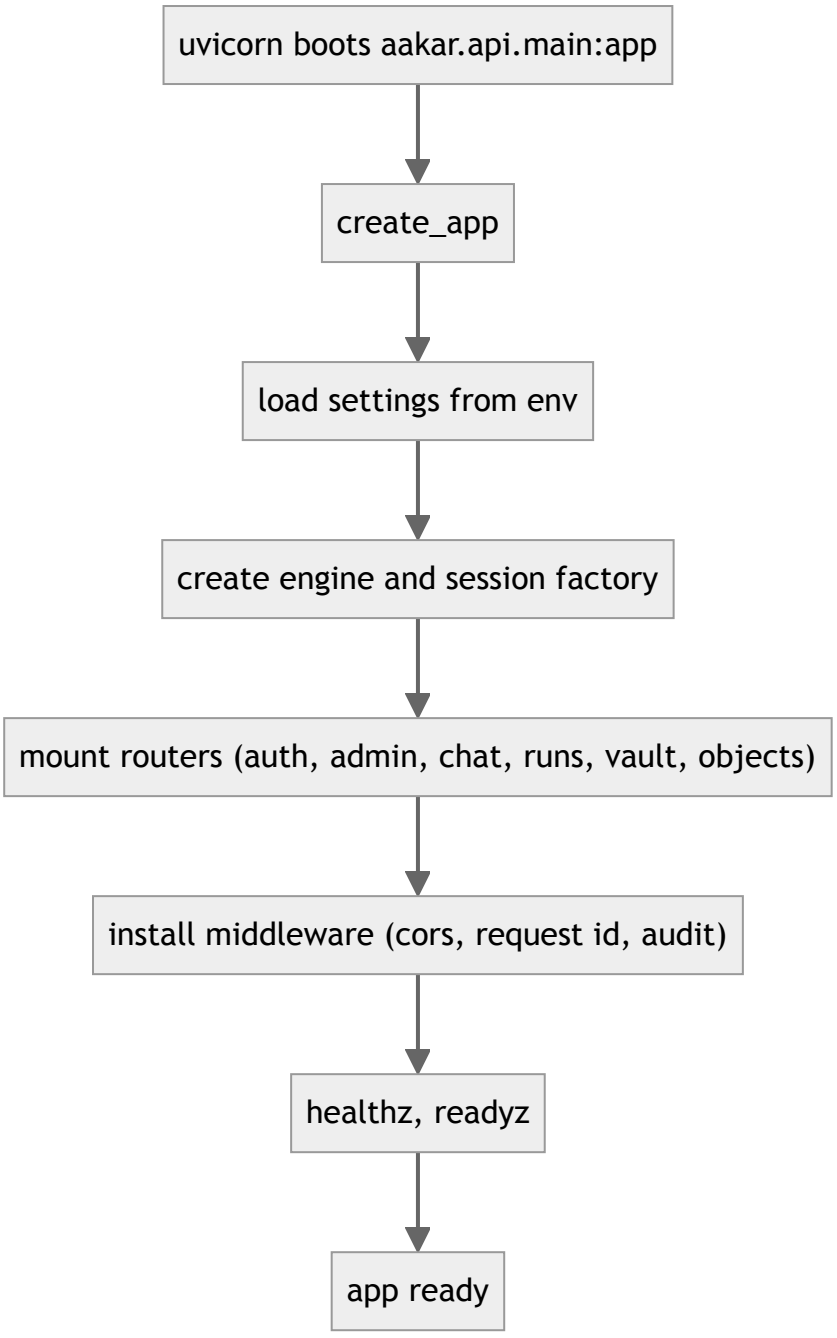
Service	Path	Port (dev)	Owners	Purpose
`aakar API`	`aakar/aakar/api`	8000	Mandala traffic	Pravesha (auth), Samvada (chat), Yajnas (runs), Kosha (vault), Bhandara (objects), admin (per-Mandala).
`admin-app API`	`admin-app/server`	8001	Demo fixture	A mock bank-ops service used as a third-party site during demos. Not the Pracharya's surface.

The split is intentional: Mandala traffic must not contend with demo traffic, and admin-app's selectors are part of the Drashtri's targeting heuristics.

2. Stack

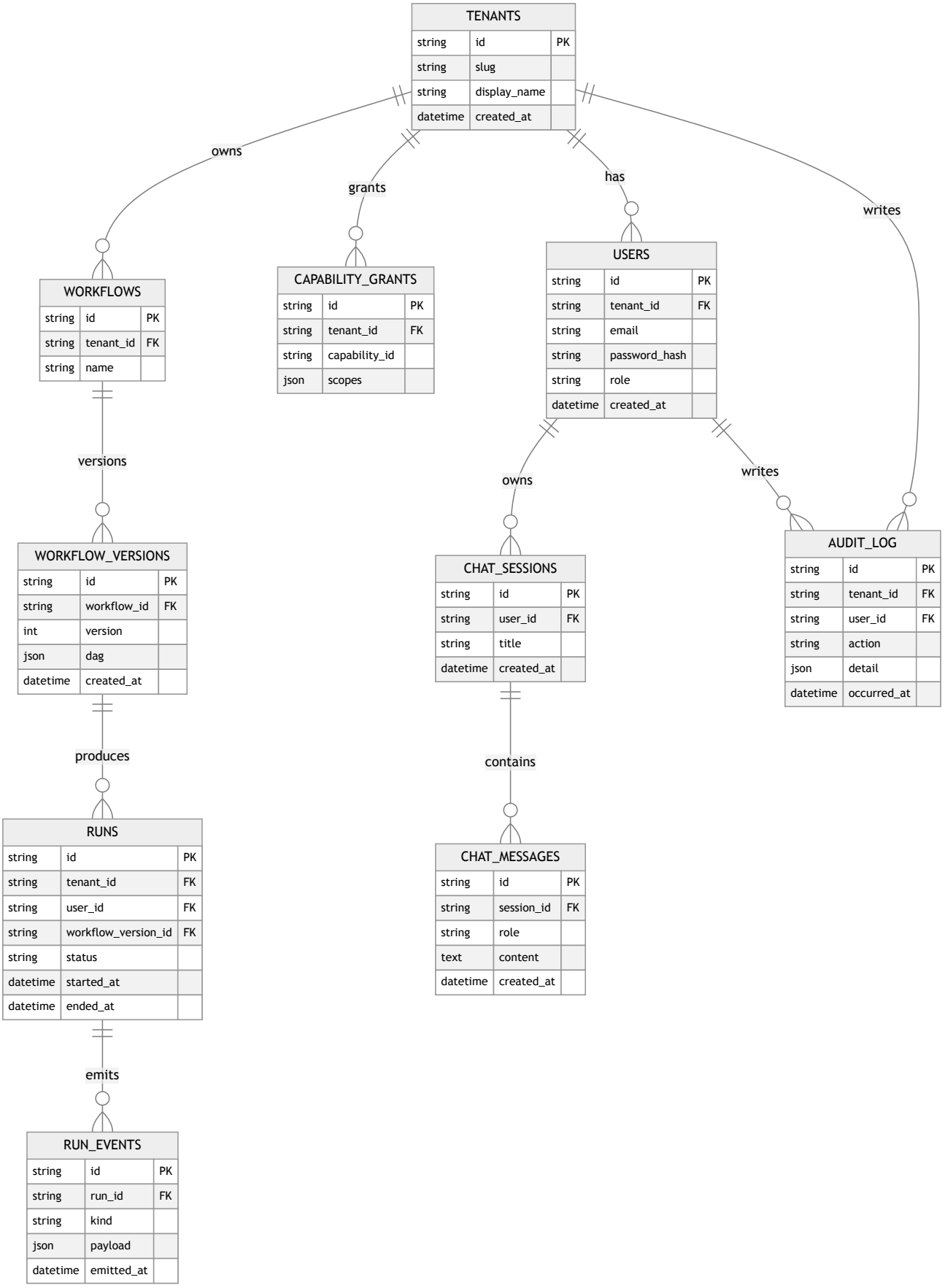
Concern	v1 choice
Language	Python 3.12
Web framework	FastAPI
Validation	Pydantic v2
ORM	SQLAlchemy 2
Migrations	Alembic
Local DB	SQLite
Cloud DB	Yugabyte (Postgres wire)
Vector index	FAISS (SQLite mode), pgvector (Yugabyte mode)
Bhandara (object store)	Filesystem under `AAKAR_DATA_DIR`
Browser worker	Playwright headless Chromium
HTTP worker	httpx
Pravesha	bcrypt + HS256 JWT
LLM (Drashtri)	OpenAI Chat Completions strict JSON
Embeddings	BGE small via sentence-transformers
Background work	asyncio + a small threadpool

3. App factory



The factory pattern keeps tests cheap: each integration test builds a fresh app bound to a temp SQLite file.

4. Database schema

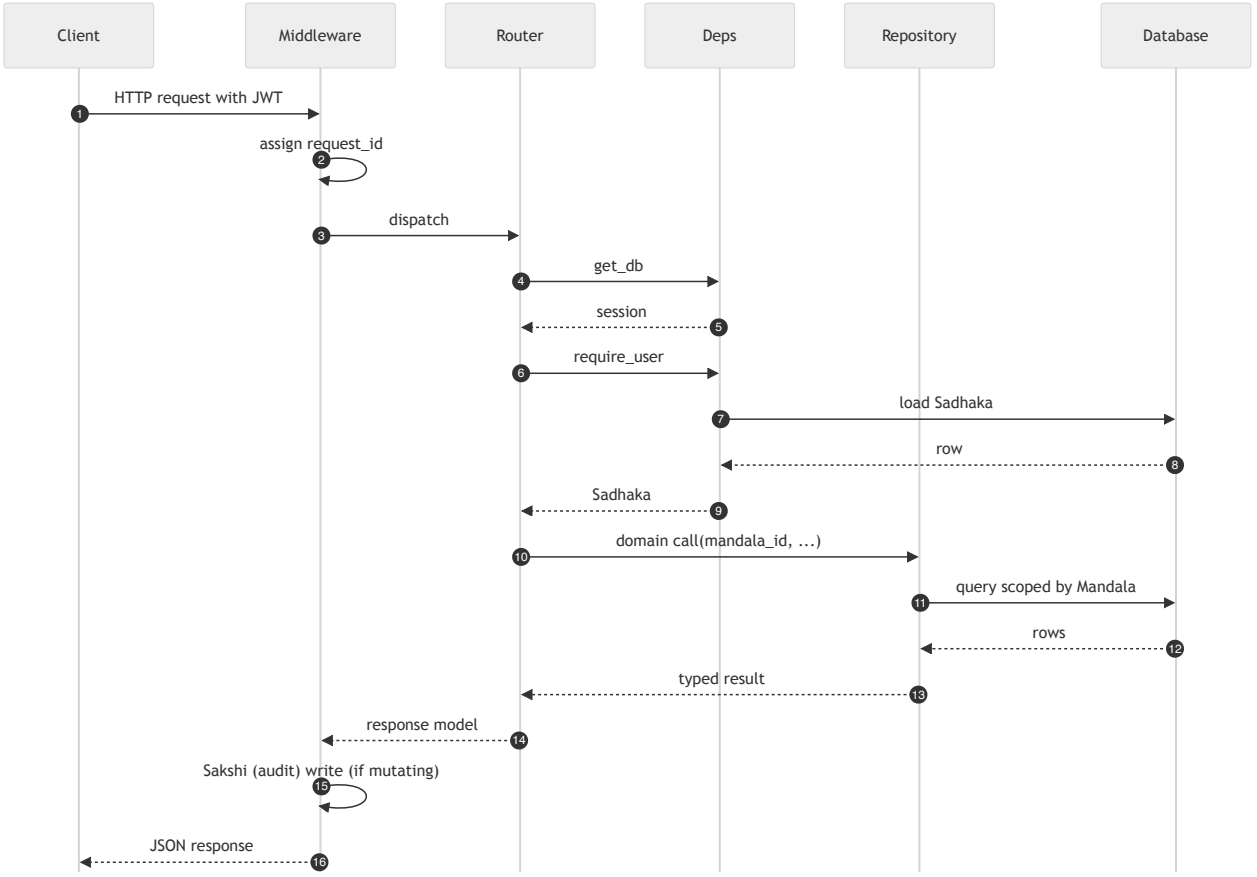


Conventions:

- All ids are ULIDs encoded as 26-char strings.

- Table names stay English (`tenants` , `users` , `runs`) for SQL clarity; the mythic names (Mandala, Sadhaka, Yajna) live in the UI and docs.
- Every Mandala-bearing table has a unique index on `(tenant_id, ...)` first; Postgres planner uses it.
- `RUN_EVENTS.payload` is JSON; PII fields are scrubbed before write.
- `AUDIT_LOG` is append-only — this table is the Sakshi (witness). Deletes are forbidden by an Alembic check.

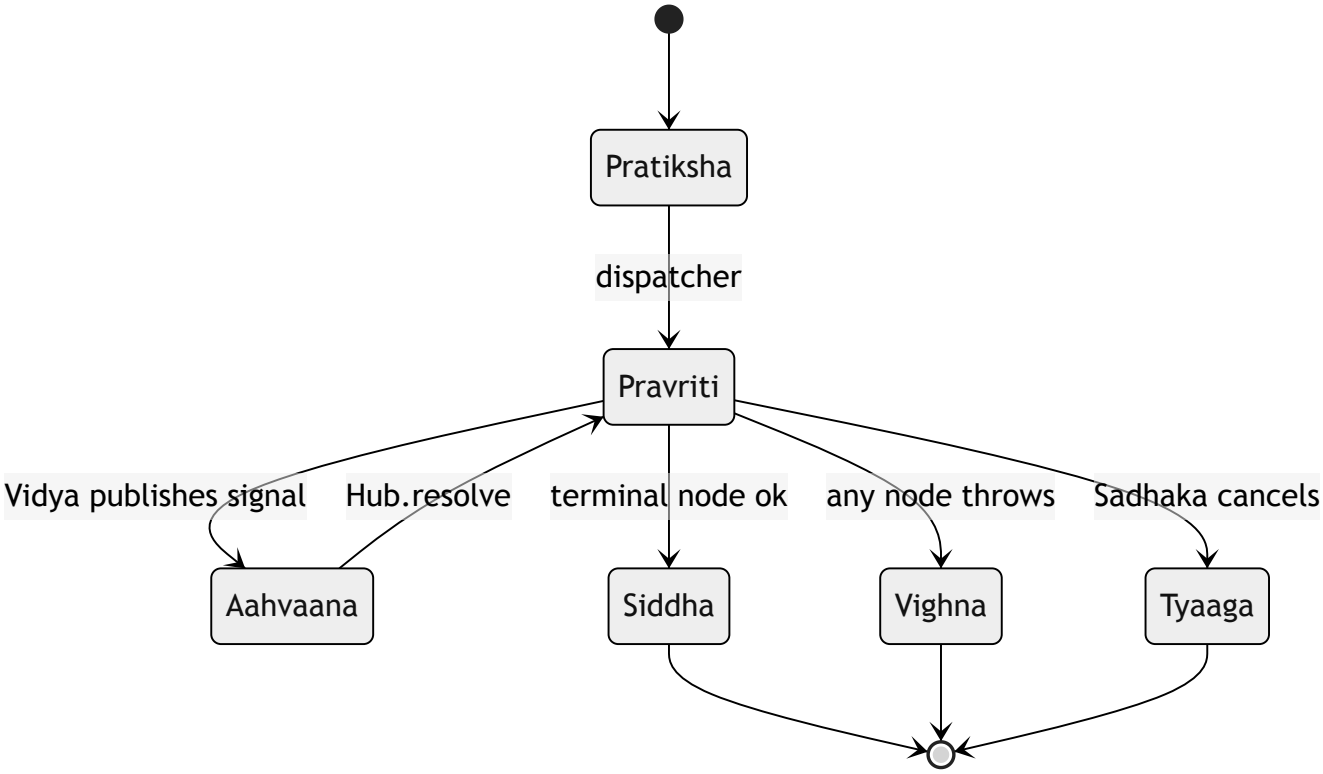
5. Request lifecycle



Mandala scoping is enforced inside repositories. Routers cannot accidentally widen the scope because repositories require a `tenant_id` argument and ignore any free-form filter that would broaden it.

6. Yajna (run) lifecycle

Mythic primary, English in parens:

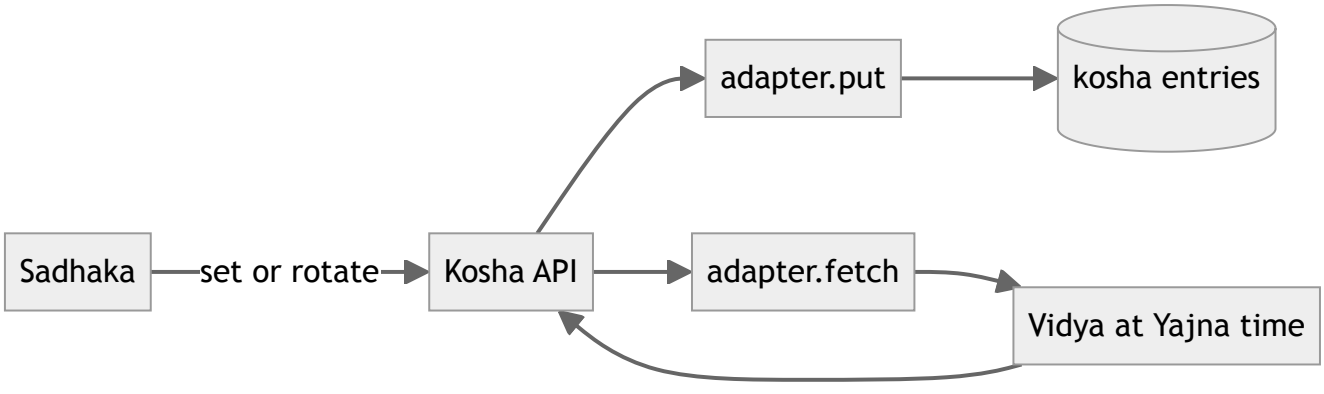


A retry of a Vighna creates a new `RUNS` row with a `parent_run_id` reference. Sakshi history is preserved.

7. Kosha (vault)

The Kosha stores per-Mandala credentials. Each entry binds:

- `mandala_id` (tenant_id on the wire)
- `site_id` (the third party, for example `nbbl` or `hdfc`)
- `user_handle` (the Sadhaka's identity on that site)
- `secret_blob` (storage form depends on adapter — see below)
- `metadata` (rotation hints, last-used)



v1 honesty: the local adapter writes JSON files at `{data}/vault/{tenant_id}/{ref}.json` with mode 0600. There is no cipher at rest in v1. A KMS-backed adapter (encrypt-on-write, decrypt-at-

read, per-Mandala data key wrapped by a master key in a managed KMS) is the Phase 2 roadmap item — code path stays the same, only the adapter changes.

8. Bhandara (object storage)

Artifacts (downloaded files, screenshots, run-time uploads) are persisted under:

```
$AAKAR_DATA_DIR/
  tenants/
    <tenant_id>/
      runs/
        <run_id>/
          events/
            <event_id>.png
          downloads/
            <filename>
          uploads/
            <filename>
```

Each row that references an artifact stores its URI as `file://...` in v1. Switching to S3 or GCS is one adapter implementation away; row-level URIs already encode the namespace.

9. FAISS index (Anveshana)

The Anveshana (Vidya search) index is built at process start from the registered Veda. It supports:

- top-k similarity search by Sankalpa embedding
- metadata filter by Mandala Adhikaras
- rebuild on Vidya registration

The index is a single FAISS `IndexFlatIP` plus a parallel sidecar list of Vidya ids. For Yugabyte deployments the same data lives in a `pgvector` table and the Drashtri uses a SQL ANN query instead. Behavior is identical from the Drashtri's perspective.

10. Darshana queries (dashboard stats)

The frontend Darshana (dashboard) pulls aggregates through three endpoints:

Endpoint	Returns
`/api/stats/runs/daily`	Yajna counts per day, last 30 days, by Avastha
`/api/stats/capabilities/usage`	top Vidyas by Yajna count
`/api/stats/sites/health`	per-site Siddhi rate over last 7 days

All three are materialized at query time from `RUNS` and `RUN_EVENTS` with simple `GROUP BY` aggregations. v1 makes no attempt at pre-aggregation.

11. Settings table

A small `settings` table holds per-Mandala feature flags and tunables:

Key	Default	Notes
`planner.strategy`	`auto`	`auto`, `oneshot`, or `agentic`.
`executor.max_concurrent_runs`	`4`	Per-Mandala cap on parallel Yajnas.
`browser.timeout_ms`	`30000`	Default per-step timeout.
`vault.rotation_days`	`90`	Reminder cadence; non-blocking.

12. Bootstrap and seeding

On first boot, the API:

1. Runs Alembic migrations to head.
2. Seeds platform metadata: Veda rows from code, Lakshana rows from YAML.
3. Optionally seeds a demo Mandala with an Acharya if `AAKAR_SEED_DEMO=1`.
4. Builds the FAISS index in memory from the seeded Vidyas.

The seed step is idempotent. Re-running it does not duplicate rows. The Pracharya (superuser) is seeded from `AAKAR_SUPERUSER_*` env vars if present; v1 deployments may run with no Pracharya at all and bootstrap one out-of-band.

13. admin-app server

The admin-app's FastAPI service is intentionally thin and **not** the Pracharya's surface:

- It owns the demo recon upload endpoint (`/api/recon/uploads`) and the upload history list — these mimic a real bank-ops backend that the Drashtri targets.
- It does not access the Mandala database. Pracharya surfaces go through the aakar API with a Pracharya JWT.
- Uploaded recon files are stored on disk under `admin-app/server/uploads/` and referenced from the in-memory history table.

14. Operations runbook (v1, dev)

Task	Command
Start everything	`./start.sh`
Run migrations	`(cd aakar && .venv/bin/alembic upgrade head)`
Run backend tests	`(cd aakar && .venv/bin/pytest)`
Reset local DB	`rm aakar/data/aakar.sqlite && alembic upgrade head`
Tail aakar logs	follow stdout in the API Terminal window
Stop a service	Ctrl-C in its window or `kill.sh`

`start.sh` opens five Terminal windows: aakar API (8000), admin-app API (8001), aakar-web (5173), admin-app (3000), nbbl-app (3001). Numerical libs are pinned to single-thread mode in the env to avoid leaked-semaphore warnings on shutdown.

15. Failure modes

Scenario	Detection	Behavior
LLM returns malformed JSON	Pydantic parse	Retry once then surface error to Samvada.
LLM emits unknown Vidya	Yantra validator	Reject Yantra, return clarification.
Browser crashes mid-Yajna	Heartbeat timeout	Mark Yajna Vighna; preserve Smritis; surface retry button.
Kosha entry missing	Vidya lookup	Pause with a `picker` Aahvaana asking the Sadhaka to add credentials.
Disk full	Bhandara write error	Mark Yajna Vighna; alert via Sakshi.
Migration failure	Alembic	Refuse to start; print the failing revision.

16. Backups (v1, manual)

Local SQLite is in `aakar/data/aakar.sqlite`; an `sqlite3 .backup` run from cron is sufficient for dev. Yugabyte is backed up via the platform's standard snapshot tooling. Bhandara contents (artifacts) are tarred per Yajna when a Yajna finishes; the tarball is sufficient evidence to replay UI state.

17. Reading guide

- For motivation and boundaries, read the HLD.
- For per-module details, read the LLD.
- For UI shape and Pratyaksha (live) panel, read the frontend doc.
- For what comes next (including OpenTelemetry), read the roadmap.